PROJECT REPORT

Air Quality & Pollution Risk Analyzer

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Air Quality & Pollution Risk Analyzer

Introduction

Air pollution is a problem for the environment and public health especially in countries like India that are growing fast. The atmosphere is filled with particulate matter (PM2.5, PM10), nitrogen oxides, and carbon monoxide which are directly linked to respiratory diseases, cardiovascular problems, and have an affect on life expectancy.

The Air Quality Index or AQI is a way to measure how polluted the air is, but it is not easily understood by the general public. Most people do not know what the numbers mean for their health or what they can do to stay safe.

This project helps people understand the AQI and what it means for their health. It uses machine learning to predict the AQI, explains the health risks and what people can do to protect themselves.

Problem Statement

Even though the AQI data is available to the public, it can be difficult to interpret. Users are often presented with numbers or labels (example: Moderate , Poor) without understanding what these categories mean for their health, how different pollutants contribute to risk, what actions they should take.

This project addresses the following problem:

How can machine learning help convert complicated air pollution data into simple health insights and practical advice for people?

Objectives

The main objectives of this project are:

- To build a machine learning model that classifies AQI levels based on the data
- To simplify complicated environmental data and make it more understandable.
- To provide information about the health risks of air pollution
- To design an interactive system for user input and real-time prediction
- To promote awareness about the impact of air pollution

Dataset

The dataset used in this project is the “Air Quality Data in India (2015–2020)” sourced from Kaggle.

It contains daily air quality readings across multiple Indian cities.

Key Features Used:

- PM2.5 (fine particulate matter)
- PM10 (coarse particulate matter)
- NO (Nitric Oxide)
- NO2 (Nitrogen Dioxide)
- CO (Carbon Monoxide)
- SO2 (Sulphur Dioxide)
- O3 (Ozone)
- Benzene
- AQI Bucket (Target variable)

Data Preprocessing

Before training the model, several preprocessing steps were performed:

- Handling Missing Values:
  Rows with missing AQI categories were removed to ensure reliable supervised learning.
- Feature Selection:
  Only relevant pollutant features were selected to reduce noise and improve model performance.
- Dropping Incomplete Rows:
  Any rows with missing pollutant values were removed.
- Label Encoding:
  AQI categories (e.g., Good, Moderate, Severe) were converted into numerical values using Label Encoding.

These steps ensure that the dataset is structured, and is suitable for machine learning.

Methodology

Choice of Algorithm

The project uses a Random Forest Classifier, which is an ensemble learning method that combines multiple decision trees.

Reasons for choosing Random Forest:

- Handles non-linear relationships effectively
- Reduces overfitting compared to single decision trees
- Works well with structured/tabular data
- Requires minimal hyperparameter tuning

Data Splitting and Scaling

The dataset was split into:

- 80% Training Data
- 20% Testing Data

The features were scaled using standard scaler to make sure they are all on the level.

Model Training Process

The model training process includes:

1. Splitting dataset into training and testing sets
2. Scaling the features
3. Training the Random Forest model
4. Evaluating accuracy using test data

The program outputs the model accuracy after training.

System Design and Workflow

The system follows a structured:

1. Data Loading
   Dataset is read and cleaned
2. Model Training
   Random Forest model is trained on processed data
3. User Input
   User enters pollutant values manually
4. Prediction
   Model predicts AQI category
5. Result Interpretation
   AQI category is mapped to:

- Health risk level
- Lung impact
- Disease risks
- Safety advice

6. Output Display
   Results are shown in a structured report format

Features of the System

This system provides several practical features:

- It can predict AQI category accurately
- Interactive user input system
- Checks to make sure input values are realistic
- Detailed health analysis:
    - Risk level
    - Effects on lungs
    - Possible diseases
    - Preventive advice
- Continuous usage loop (user can check multiple readings)

Results and Observations

The model successfully classifies AQI categories based on pollutant levels.

Observations:

- Higher PM2.5 and PM10 levels make the air quality worse
- The model performs well because the data is clean and structures
- The addition of health insights makes the system more useful than a standard classifier

The output is not just a prediction, but a complete risk report, making it practical for everyday use.

Key Design Decisions

Several important decisions were made during development:

- Using Random Forest instead of simpler models:

  To achieve better accuracy and robustness

- Limiting features to major pollutants:

  To reduce complexity and improve interpretability

- Adding health effects mapping:

  To make predictions meaningful for users

- Using a CLI-based interface:

  To keep it simple and focus on logic

Challenges Faced

During the development of this project, the following challenges were encountered:

- Handling missing and inconsistent data
- Selecting the most relevant features
- Converting technical AQI outputs into meaningful health insights
- Designing user input validation for realistic pollutant ranges
- Ensuring model accuracy without overfitting

Applications

This project has many real-world applications:

- Public awareness platforms
- Health advisory systems
- Smart city environmental monitoring
- Integration into mobile or web apps
- Educational tools for understanding air pollution

Limitations

Despite its usefulness, the system has some limitations:

- It is a CLI-based application (no graphical interface)
- Depends on input data

Does not use real-time pollution data

- Limited to the features available in the dataset
- Model performance depends on dataset quality

Future Enhancements

The system can be improved further by:

- Developing a web or mobile application interface
- Integrating real-time AQI APIs
- Adding graphical data visualization
- Using advanced models like Gradient Boosting or Neural Networks
- Including location-based predictions
- Expanding dataset coverage

Conclusion

This project shows how machine learning can be used in a practical way to address real-world environmental problems like air pollution. Instead of just working with raw data, it focuses on turning that data into something people can actually understand and use.

By combining AQI prediction with health risk analysis, the system not only identifies pollution levels but also explains what those levels mean for human health. This makes the output more meaningful and easier to interpret.

As a result, the project is not just a technical implementation of a machine learning model, but also a useful tool that can help raise awareness about air pollution and encourage people to take informed precautions in their daily lives.

References

- Air Quality Data in India (Kaggle)
- Scikit-learn Documentation
- AQI India Guidelines